

HSiKAN: Mathematical Foundation

Eight layers, eight decades

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HSiKAN (Hypergraph Signed Kolmogorov–Arnold Network) is the composition of eight mathematical layers, each pulled from a different established literature spanning 1946 to 2024. This document maps the layers in dependency order, naming the source results and the role each plays inside the model.

1. Signed-graph theory

Heider 1946 → Cartwright & Harary 1956 → Davis 1967

- Signed graph $G = (V, E, s)$ with $s : E \rightarrow \{+1, -1\}$.
- **Cycle balance theorem.** A cycle c is *balanced* iff

$$\prod_{e \in c} s(e) = +1.$$

- **Structural balance (Cartwright–Harary 1956).** A complete signed graph is balanced iff V partitions into two mutually-hostile cliques. Davis 1967 generalises to k -clusterability (weak balance: no all-negative triangle).
- **HSiKAN’s σ -rule.** The per-vertex sign within a cycle is the cycle-product specialised to vertex i :

$$\sigma_i(c) = (-1)^{n_i^-(c)},$$

where $n_i^-(c)$ is the count of negative edges incident to vertex i within cycle c .

2. Hypergraph theory

Berge 1973; Thesis 2 (this dissertation)

- A k -uniform hypergraph generalises edges to k -vertex relations.
- Two canonical reductions to ordinary graphs:
 - **Star expansion.** Hyperedge $e \mapsto$ centroid vertex c_e plus k bipartite edges $\{(v_i, c_e) : v_i \in e\}$. The result is bipartite $\Rightarrow$ only even- k cycles exist in the expansion. *This is HSiKAN’s input format.*
 - **Clique expansion.** k -edge $\mapsto \binom{k}{2}$ pairwise edges. Discards arity information.
- Sparse signed-incidence buffers enter at training:

$$\mathbf{M}_e^{(k)} \in \mathbb{R}^{|E_{\text{test}}| \times |T_k|} \quad \text{and} \quad \mathbf{M}_{vt} \in \mathbb{R}^{|V| \times |T|}.$$

3. Kolmogorov–Arnold representation

Kolmogorov 1957 → Liu et al. 2024 (KAN)

- **KA superposition theorem.** Any continuous $f : [0, 1]^n \rightarrow \mathbb{R}$ admits the decomposition

$$f(x) = \sum_{q=0}^{2n} \Phi_q \left(\sum_{p=1}^n \psi_{q,p}(x_p) \right)$$

with $2n+1$ outer and $n(2n+1)$ inner univariate functions.

- **kans** parameterise Φ_q and $\psi_{q,p}$ as splines.
- **KA-rank ladder for signed graphs.** Three points on the spectrum:
 - rank-1: HGNN (linear superposition);
 - rank-2: Signed-HGNN;
 - full rank: HSiKAN (per-sign nonlinear inner+outer splines).

4. Spline approximation

Catmull–Rom 1974, de Boor 1978, Kochanek–Bartels 1984

- Inner φ_v^σ and outer φ_e^σ are uniform Catmull–Rom cubic splines on $[-1, 1]$:
 - 4-point gather ($P_{i-1}, P_i, P_{i+1}, P_{i+2}$) per output;
 - constant 4×4 basis matrix M_{CR} folds the Hermite blend coefficients;
 - C^1 -continuous, interpolatory — the curve passes through the control points.
- Backbone-invariant: B-splines (Cox–de Boor recursion) and Kochanek–Bartels (TCB tension / continuity / bias parameters, reducing to Catmull–Rom at TCB = (0, 0, 0)) yield the same downstream AUCs to within seed noise.

5. Mixed-arity mixing

The per-arity cycle embeddings are combined by a softmax-normalised learnable weight vector α :

$$\mathbf{H}^{\text{edge}} = \sum_{k \in \mathcal{K}} \text{softmax}(\alpha)_k \cdot \mathbf{M}_e^{(k)} \mathbf{H}^{(e,k)}.$$

α_k acts as a **quantitative ethogram-like read-out**: which cycle order does this dataset’s signed signal live at? Empirically: Bitcoin $\rightarrow k=3$ triads, Slashdot $\rightarrow k=5$ (46%), SBM $\rightarrow k=4$ matching its 4-community design.

6. Symbolic distillation (Fourier-style)

After training, each surviving spline is fit against a six-form symbolic library

$$\{\text{linear, quadratic, cubic, } a \sin(\omega x + \varphi) + c, \exp, \tanh\}$$

via Levenberg–Marquardt least squares. Approximately 91% of survivors fit best as a sinusoid. Validated on Bitcoin Alpha + OTC (3 seeds, $h=16$) against three null baselines:

baseline	sinusoidal fraction
trained HSiKAN	0.901 ± 0.055
untrained HSiKAN	0.505 ± 0.036
random spline coefs	0.583 ± 0.045
Gaussian-process draws	0.499 ± 0.021

The +35–40 pp gap rules out spline-basis bias and grid-sampling artefacts.

7. Cycle enumeration (algorithmic)

- DFS with `Vec<u64>` bitset visited ($8\times$ smaller than `Vec<bool>`, fits in L1 for graphs up to $\sim 500k$ nodes).
- BFS-distance pruning to the start vertex (skip branches that cannot close in the remaining budget).
- **Color-coding (Alon–Yuster–Zwick 1995).** Random k -colouring of vertices; only rainbow cycles are emitted; per-colouring cost is proportional to $|\{\text{rainbow cycles}\}|/|\{\text{total cycles}\}|$. Probability a fixed k -cycle is rainbow $= k!/k^k$.
- Vitter reservoir sampling for unbiased cycle subsets.
- Canonical form: smallest-vertex rotation + lex orientation tiebreak.
- **Walk-HSiKAN variant.** Same DFS, no closure check; open length- L simple walks emitted with $\text{path}[0] \leq \text{path}[L]$ canonicalisation.
- **Performance.** Slashdot $k=4$ in 4.6 s for 100 k cycles via Rust + rayon + atomic early-stop, vs ~ 227 s for the equivalent serial Python.

8. Information geometry and regularisation

- **Spectral entropy** as a KA-bound:

$$H(W) = - \sum_i \tilde{\lambda}_i \log \tilde{\lambda}_i / \log \text{rank}(W).$$

- **Sanchez–Giraldo matrix-based Rényi- α entropy** estimator (kernel α -norm).
- **Multi-term Lyapunov wrapping:**

$$\lambda_{\text{eff}} = \lambda_0 \exp(-\eta D_{\text{KL}}(\hat{\lambda}^{(t-1)} \parallel \hat{\lambda}^{(t)})).$$

- **Path I calibration law:** $\lambda_{\text{multi}} \sim \lambda_{\text{scalar}}/L$ — depth-dependent rescaling. Empirically validated by ph18c: highway-10 confirmed at $\lambda=1.0$ (+0.092 pp, $p < 0.05$); resmlp-40 ($L=40$) recovers at $\lambda=0.5$ (+0.213 pp, $p < 0.05$), the predicted ratio $\approx 1/4$ of the $L=10$ value.

What today’s chicken-aggression work added

- **Layer 1 reproduces itself in a flock simulator.** The Cartwright–Harary structural-balance theorem implies aggressors generate *balanced* triads: the (aggressor, v_1, v_2) pattern has signs $(-, -, +)$ because two victims of the same aggressor are typically peaceful with each other. Sign product $= +1$, so the cycle is balanced. Our unsupervised cycle-balance scorer reads off per-vertex aggressor probability directly from this fraction — *graph theory from 1956 identifying which chicken is the aggressor with zero labels*.

- **Layer 2 generalises to anatomy.** The chicken body itself is a hypergraph: 12 keypoints, rigid bones (sign $+1$), flexible joints (sign -1), kinematic chains as multi-vertex hyperedges. Star-expandable through the same machinery. Per-bird anatomy nests inside per-flock interaction — a multi-scale signed-hypergraph description of an entire flock.

The aesthetic point

The stack composes from **1946 (Heider)** $\rightarrow$ **1956 (Cartwright–Harary)** $\rightarrow$ **1957 (Kolmogorov)** $\rightarrow$ **1973 (Berge)** $\rightarrow$ **1974 (Catmull–Rom)** $\rightarrow$ **1995 (Alon–Yuster–Zwick)** $\rightarrow$ **2024 (Liu kan)** $\rightarrow$ **2026 (HSiKAN)**.

Eighty years of pure mathematics, quietly waiting to read off which chicken is the aggressor.